Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 1 of 7 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

5(a)	Gradient $PR = \frac{6}{-6} = -1$, Gradient $QS = \frac{8}{8} = 1$	M1A1
	Product $= -1 \Rightarrow$ perpendicular	A1 (3)
(b)	(i) $PR = \sqrt{6^2+6^2} = 6\sqrt{2}$ $\left(=\sqrt{72}\right)$	M1
	(ii) $QS = \sqrt{8^2+8^2} = 8\sqrt{2}$ $\left(=\sqrt{128}\right)$	A1 (2)
(c)	Area $= \frac{1}{2}$ "$6\sqrt{2}$"×"$8\sqrt{2}$" $= 48$ $\left(\text{units}^2\right)$	M1A1 (2)
		[7]

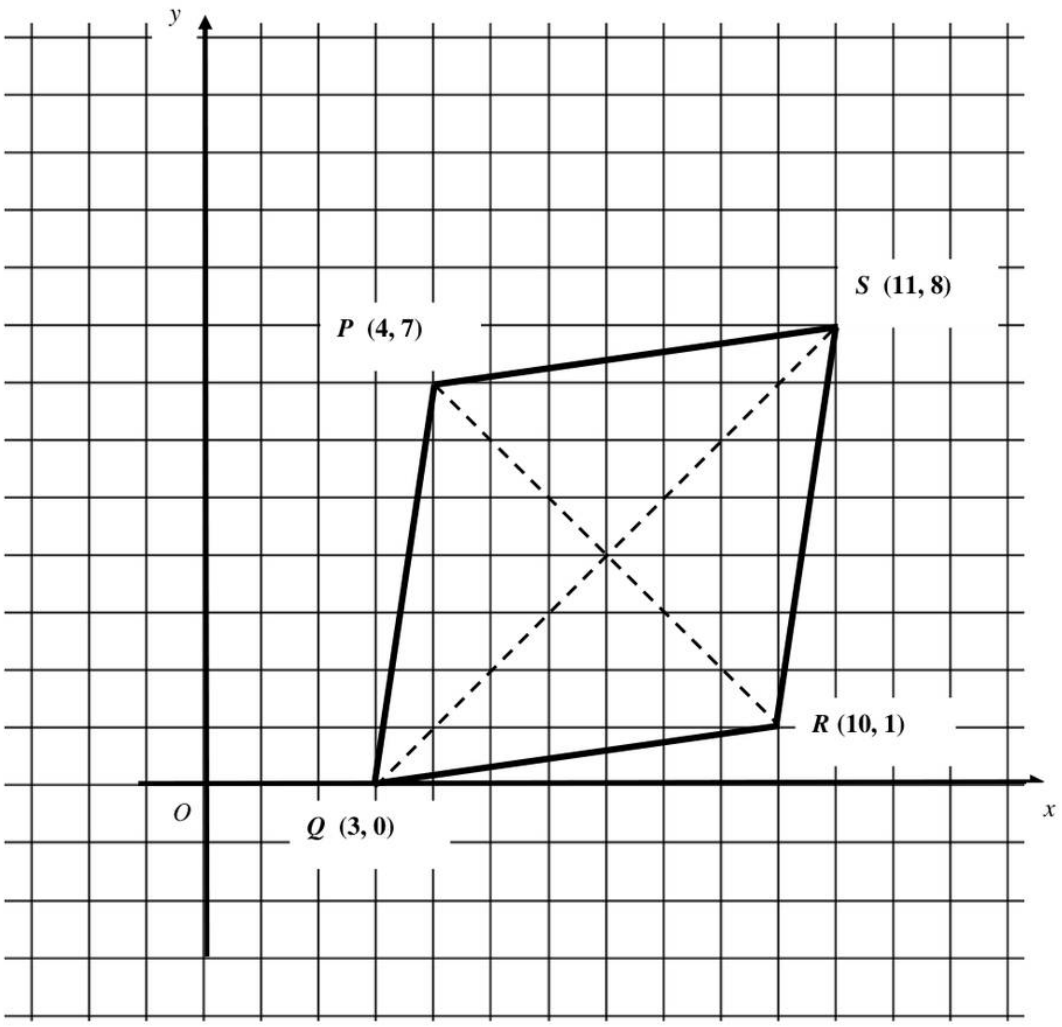
Part	Mark	Notes
(a)	M1	Finds the gradient of PR and QS using a correct method. This may be on a diagram. Gradient $PR = \frac{7-1}{4-10} = \frac{6}{-6} = -1$, Gradient $QS = \frac{8-0}{11-3} = \frac{8}{8} = 1$
	A1	Both gradients correct Gradient $PR = -1$, Gradient $QS = 1$
	A1	Finds the product of the two gradients with a statements that as the product = –1 then the lines are perpendicular.
(b)	M1	For **either** $PR = \sqrt{6^2+6^2} = \sqrt{72}$ or $6\sqrt{2}$ **OR** $QS = \sqrt{8^2+8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
	A1	For **both** $PR = \sqrt{6^2+6^2} = \sqrt{72}$ or $6\sqrt{2}$ **AND** $QS = \sqrt{8^2+8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
(c)	M1	$PQRS$ is a kite so $= \frac{1}{2}$ "$6\sqrt{2}$"×"$8\sqrt{2}$" $= \ldots$
	A1	Area = 48 $\left(\text{units}^2\right)$
	ALT Uses determinants	
	M1	Area $= \frac{1}{2}\begin{pmatrix} 4 & 3 & 10 & 11 & 4 \\ 7 & 0 & 1 & 8 & 7 \end{pmatrix}$ $= \frac{1}{2}\left(\left[4\times0+3\times1+10\times8+11\times7\right]-\left[3\times7+10\times0+11\times1+4\times8\right]\right)$ $= \ldots$ Allow one slip in a product.
	A1	Area = 48 $\left(\text{units}^2\right)$

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 1 of 7 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

USEFUL SKETCH



Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 1 of 7 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 1 of 7 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 2 of 7 · Question 10 part c.i · 3 marks

2025 NOV · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

Question	Scheme	Marks
10 (a)	$10-3a+a=0 \Rightarrow 10=2a \Rightarrow a=5$ *	M1A1cso [2]
(b)	$(0,-5)$	B1 [1]
(c) (i)	$A \quad (5,10)$ $C \quad (50,-5)$ $y=3x-5 \Rightarrow m_{L1}=3 \qquad m_{L2}=\dfrac{-5-10}{50-5}=\dfrac{-1}{3}$ $m_{L2}\times m_{L1}=\dfrac{-1}{3}\times 3=-1$	B1 M1A1 [3]
(ii)	$y-(-5)=\dfrac{-1}{3}(x-50) \Rightarrow y=\dfrac{-x}{3}+\dfrac{35}{3}$	M1A1 [2]
(d)	$-3=\dfrac{n-("-5")}{m} \Rightarrow n=-5-3m$ $2250=(50-m)^2+(-5-n)^2$ $2250=(50-m)^2+(-5-(-5-3m))^2=(50-m)^2+(3m)^2$ $2250=2500-100m+10m^2 \Rightarrow 0=25-10m+m^2=(m-5)^2$ $\Rightarrow m=5$ $\Rightarrow n=-5-3\times 5=-20$ **ALT** $-3=\dfrac{n-("-5")}{m} \Rightarrow m=\dfrac{-(n+5)}{3}$ $2250=(50-m)^2+(-5-n)^2$ $2250=\left(50+\dfrac{n+5}{3}\right)^2+n^2+10n+25$ $2250=2500+\dfrac{100(n+5)}{3}+\dfrac{n^2+10n+25}{9}+n^2+10n+25$ $\Rightarrow \quad 0=9(2500-2250)+300(n+5)+10n^2+100n+250$ $=2500+300n+1500+10n^2+100n$ $=n^2+40n+400$ $=(n+20)(n+20)$ $\Rightarrow n=-20$ $\Rightarrow m=\dfrac{-(-20+5)}{3}=5$	M1 M1 M1 M1 A1 A1 [6] [M1] [M1] [M1] [M1] [A1] [A1]
(e)	Correctly uses the formula for a kite using their lengths	M1

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 2 of 7 · Question 10 part c.i · 3 marks

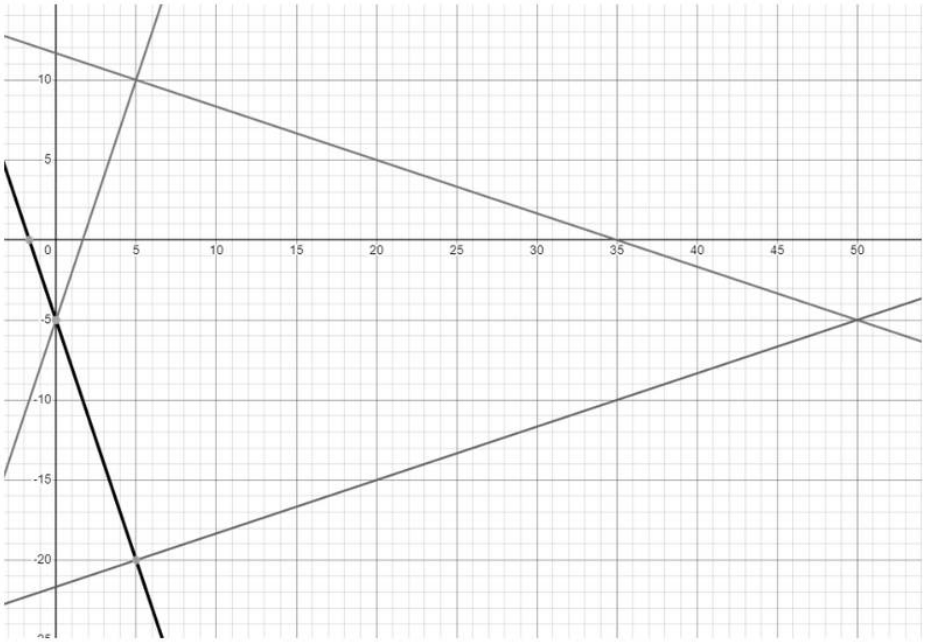
2025 NOV · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

	$\text{Area} = \frac{50\times(10-(-20))}{2} = 750$	dM1A1 [3]
	ALT 1 Correctly uses determinant method with their co-ordinates $\text{Area} = \frac{1}{2}\begin{pmatrix} 0 & 5 & 50 & 5 & 0 \\ -5 & 10 & -5 & -20 & -5 \end{pmatrix} = 750$	[M1 dM1A1]
	ALT 2 Correctly uses formulae for two triangles ABD and ACD using their lengths $\text{Area} = 2\times\left(\frac{1}{2}\times 15\times 50\right) = 750$ Or $\text{Area} = \left(\frac{1}{2}\times 30\times 5\right)+\left(\frac{1}{2}\times 45\times 30\right) = 750$	[M1 dM1A1]
		Total 17 marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 2 of 7 · Question 10 part c.i · 3 marks

2025 NOV · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract



Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 2 of 7 · Question 10 part c.i · 3 marks

2025 NOV · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

Question	Scheme	Marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 3 of 7 · Question 4 part d.ii · 4 marks

2021 JUN · Paper 02 · extract 1 of 5 · find part d.ii in this mark scheme extract

number		
4 (a)	Gradient $= \dfrac{11+10}{3+4} = 3$	
	$y+10 = 3(x+4)$ or $y-11 = 3(x-3)$ oe	M1A1 (2)
(b)	e.g. $\left(\dfrac{4\times-4+3\times3}{3+4}, \dfrac{4\times-10+3\times11}{3+4}\right) = (-1,-1)$	M1 A1 (2)
ALT (b)	Using Vectors	
	$\begin{pmatrix}-4\\-10\end{pmatrix}+\dfrac{3}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$ or $\begin{pmatrix}3\\11\end{pmatrix}-\dfrac{4}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$	{M1} {A1}
(c)	$-\dfrac{1}{3}=\dfrac{n+1}{m+1} \Rightarrow -\dfrac{1}{3}(m+1)=n+1$	M1
	$\left(\sqrt{10}\right)^2=(m+1)^2+(n+1)^2$	M1
	$10=(m+1)^2+\dfrac{1}{9}(m+1)^2$	M1
	$9=(m+1)^2$	M1
	$m=-4 \quad n=0$	A1 A1 (6)
ALT (c)	Using Vectors	
	$\overrightarrow{AB}=\begin{pmatrix}7\\21\end{pmatrix}$ so perpendicular to $\overrightarrow{AB}=\begin{pmatrix}21\\-7\end{pmatrix}$	{M1}
	$\left\lvert\overrightarrow{AB}\right\rvert = 7\sqrt{10} \Rightarrow, \left\lvert\overrightarrow{AP}\right\rvert = 3\sqrt{10}$	{M1,M1}
	$\overrightarrow{PQ}=\dfrac{\sqrt{10}}{7\sqrt{10}}\times\begin{pmatrix}21\\-7\end{pmatrix}=\begin{pmatrix}3\\-1\end{pmatrix}$	{M1}
	So $Q=(-1-3,-1--1)$	{A1}
	$Q=(-4,0)$	{A1}
(d)(i)	$AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$	M1
	$RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$	A1
(d)(ii)	Gradient of $RQ=\dfrac{-21-0}{-11+4}=3$	M1
	So Gradient of AB (=3) = Gradient of RQ	A1 (4)

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 3 of 7 · Question 4 part d.ii · 4 marks

2021 JUN · Paper 02 · extract 2 of 5 · find part d.ii in this mark scheme extract

ALT (d)	Using Vectors $\overrightarrow{RQ} = \begin{pmatrix} -4-(-11) \\ 0-(-21) \end{pmatrix} = \begin{pmatrix} 7 \\ 21 \end{pmatrix}$	{M1} {A1}
	$\overrightarrow{AB} = \begin{pmatrix} 7 \\ 21 \end{pmatrix} = \overrightarrow{RQ}$	{M1}
	Because the vectors are the same they must be parallel and the same length	{A1}
(e)	Area = $7\sqrt{10} \times \sqrt{10} = 70$	M1 A1 (2)
ALT (e)	Using Vectors $\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & -4 & 3 \\ 11 & 10 & -21 & 0 & 11 \end{vmatrix}$	{M1}
	$= 70$	{A1}
		Total is 16 marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 3 of 7 · Question 4 part d.ii · 4 marks

2021 JUN · Paper 02 · extract 3 of 5 · find part d.ii in this mark scheme extract

Part	Mark	Guidance
(a)	M1	For a fully correct method of finding an equation of a straight line. $\frac{y-y_1}{y_2-y_1}=\frac{x-x_1}{x_2-x_1}\Rightarrow\frac{y-(-10)}{11-(-10)}=\frac{x-(-4)}{3-(-4)}$ Or finds gradient $\frac{11+10}{3+4}=3$ and uses $y+10=3(x+4)$ or $y-11=3(x-3)$ If $y=mx+c$ is used, they must find a complete equation for this mark. Allow **one** error only for the award of this mark.
	A1	For a correct line in any form. $y+10=3(x+4)$ or $y-11=3(x-3)$ or $y=3x+2$ or even $\frac{y+10}{21}=\frac{x+4}{7}$ but do not allow incomplete processing.
(b)	M1	For one correct from $x=-1$ or $y=-1$
	A1	For the correct coordinates of point P $(-1,-1)$ Accept $x=-1$ $y=-1$
(c)	M1	Uses the perpendicular gradient to set up an equation in m and n. $-\frac{1}{'3'}=\frac{n-'(-1)'}{m-'(-1)'}\Rightarrow-\frac{1}{'3'}(m+1)=n+1$ or $n=-\frac{1}{3}m-\frac{4}{3}$ Ft their gradient in part (a) and their P from part (b) for this mark.
	M1	Uses Pythagoras theorem to set up an equation in m and n. $\left(\sqrt{10}\right)^2=(m-'(-1)')^2+(n-'(-1)')^2$ Ft their coordinates of point P form part (b) for this mark.
	M1	Attempts to solve their two equations in n and m simultaneously and forms a quadratic equation in one variable only. $10=(m+1)^2+\frac{1}{9}(m+1)^2\Rightarrow 9=(m+1)^2$ or $0=m^2+2m-8$ or $10=9(n+1)^2+(n+1)^2\Rightarrow 0=10n^2+20n$
	M1	For solving their either: $9=(m+1)^2\Rightarrow m=...$ or $0=10n^2+20n\Rightarrow n=...$ which must be a quadratic equation.
	A1	For finding **either** $m=-4$ **or** $n=0$ Condone the sight of $m=2$ for this mark.
	A1	For finding **both** $m=-4$ **and** $n=0$ $\Rightarrow(-4,\ 0)$ The final answer must be given as coordinates.
	ALT – using vectors – see main scheme.	
(d)(i)	M1	For finding **either** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **Or** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$
	A1	For finding **both** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **And** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$ **and** states they are equal
(d)(ii)	M1	The gradient of $RQ=\frac{-21-'0'}{-11-'(-4)'}='3'$ Ft their coordinates from part (c)

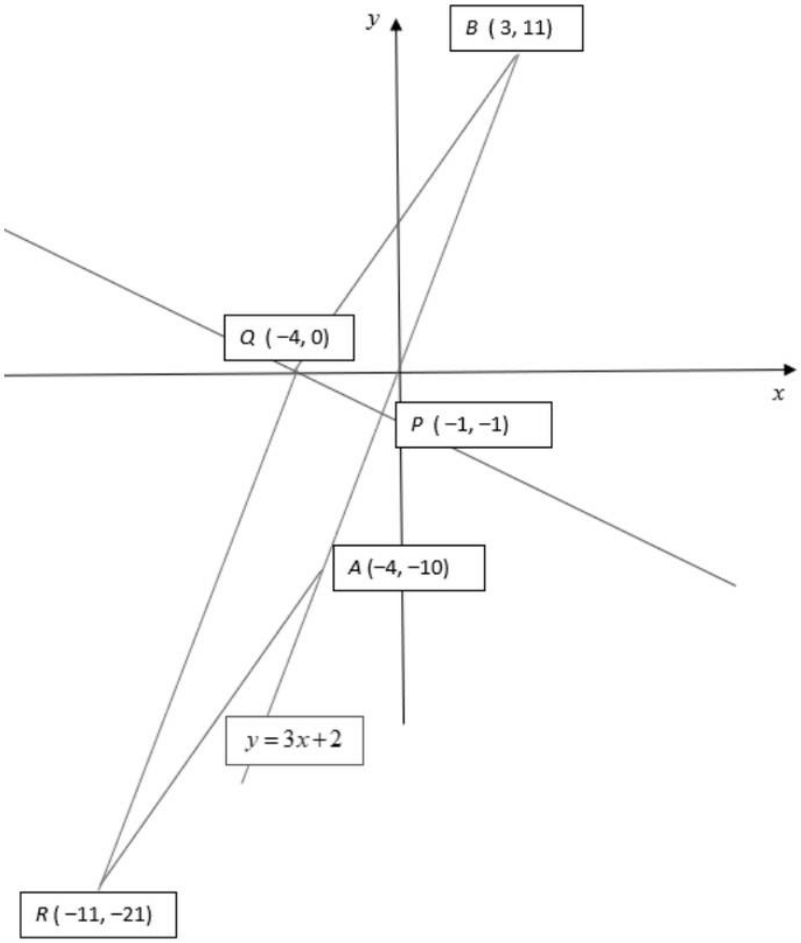
Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 3 of 7 · Question 4 part d.ii · 4 marks

2021 JUN · Paper 02 · extract 4 of 5 · find part d.ii in this mark scheme extract

	A1	States that the gradient of RQ = gradient of AB [from (a)]
	ALT – Uses vectors, see main scheme. Ft their coordinates of $Q - (m, n)$	
(e)	**M1**	For a correct expression for the area using their length of AB and the given length of PQ $\left(\sqrt{10}\right)$ Area = $'7\sqrt{10}' \times \sqrt{10} = ...$
	A1	For the area = 70 [square units]
	ALT – Uses the discriminant	
	M1	For a correct expression of the area in sequential order using their coordinates for Q Area = $\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & '-4' & 3 \\ 11 & 10 & -21 & '0' & 11 \end{vmatrix}$
	A1	Area = 70 [square units]

Useful sketch



Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 3 of 7 · Question 4 part d.ii · 4 marks

2021 JUN · Paper 02 · extract 5 of 5 · find part d.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 4 of 7 · Question 2 part c · 3 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

2 (a)	Gradient $= \frac{4+1}{3+7} = \frac{1}{2}$	M1
	$y+1 = \frac{1}{2}(x+7)$	M1
	$x-2y+5=0$	A1 (3)
(b)	$AB = \sqrt{10^2+5^2} = 5\sqrt{5}$ $AC = \sqrt{(-7--3)^2+(7--1)^2} = 4\sqrt{5}$	M1
	$k = \frac{5}{4}$	A1 (2)
(c)	$-2 = \frac{7-p}{-3-3}$	M1
	$12 = 7-p$	dM1
	$p=-5$	A1 (3)
		Total 8 marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 4 of 7 · Question 2 part c · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 5 of 7 · Question 11 part b · 7 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Number		
11 (a)	$\frac{y-5}{5--5}=\frac{x-7}{7--3}$ $[\Rightarrow y=x-2]$ oe	M1A1 [2]
(b)	$m_{BC}=\frac{5-1}{7-p}=\frac{4}{7-p}$, $m_{AC}=\frac{1-(-5)}{p-(-3)}=\frac{6}{p+3}$ $m_{BC}\times m_{AC}=-1\Rightarrow\frac{4}{7-p}\times\frac{6}{p+3}=-1\Rightarrow\frac{4}{7-p}=-\left(\frac{p+3}{6}\right)$ $\Rightarrow -24=-p^2+4p+21\Rightarrow p^2-4p-45=0$ $\Rightarrow (p-9)(p+5)=0\Rightarrow p=-5,\ (9)$	M1A1A1 M1 dM1 M1A1* [7]
(c)	Using formula; $\left(\frac{(7\times-5)-(3\times7)}{7-3},\frac{(7\times1)-(3\times5)}{7-3}\right)=(-14,-2)$	B1B1 [2]
(d) (i)	$AC=\sqrt{(1-[-5])^2+(-5-[-3])^2}=\sqrt{40}=2\sqrt{10}$	B1
(ii)	$BD=\sqrt{(7-[-14])^2+(5-[-2])^2}=\sqrt{490}=7\sqrt{10}$ Area $=\frac{1}{2}\times2\sqrt{10}\times7\sqrt{10}=70\ (\text{units}^2)$	M1 M1A1 [4]
		Total 15 marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 5 of 7 · Question 11 part b · 7 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For forming an equation of a straight line
	A1	Accept a correct line in any form following simplification of denominators; for example, accept $\frac{y-5}{10}=\frac{x-7}{10}$ or $y-5=x-7$ or $y+5=x+3$
(b)	M1	For attempting to find an expression for the gradient of BC **or** the gradient of AC $m_{BC}=\frac{5-1}{7-p}$ or $m_{AC}=\frac{1-(-5)}{p-(-3)}$ Accept maximum one slip in either expression for this mark.
	A1	For an expression for the gradient of BC **or** AC fully correct
	A1	For an expression for the gradient of BC **and** AC fully correct Need not be simplified in either case
	M1	For setting their gradient of BC = negative reciprocal of their gradient of AC
	dM1	For attempting to form a 3TQ in p: $\left[p^2-4p-45=0\right]$ Dependent on the previous M mark.
	M1	For solving their 3TQ by any method. See General Guidance.
	A1*	For $p=-5$
(c)	B1	For either $x=-14$ or $y=-2$
	B1	Both coordinates correct $(-14,-2)$ given as coordinates only.
(d)(i)	B1	For the correct length of AC $\left[=2\sqrt{10}\right]$
(ii)	M1	For the correct method to find length BD $\left[=7\sqrt{10}\right]$ Ft their $x=-14$ or $y=-2$
	M1	For a correct method using $A=\frac{1}{2}\times\text{base}\times\text{height}$ using their AC and BD
	A1	For the correct area of $70\ \left(\text{units}^2\right)$
	ALT – Uses determinants	
	M1	For stating $A=\frac{1}{2}\begin{pmatrix}-3 & 7 & -14 & -3\\ -5 & 5 & -2 & -5\end{pmatrix}$ oe
	M1	For evaluating their determinant $A=\frac{1}{2}\left(\left[-3\times5+7\times-2+-14\times-5\right]-\left[7\times-5+-14\times5+-3\times-2\right]\right)$ oe
	A1	Area = $70\ \left(\text{units}^2\right)$ Which must be a positive value

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 5 of 7 · Question 11 part b · 7 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

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Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 6 of 7 · Question 2 part c · 8 marks

2025 JUN · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

2(a)	$\sqrt{(8-3)^2+(3-2)^2}$ or $\sqrt{(4-3)^2+(7-2)^2}$ or $\sqrt{(8-4)^2+(3-7)^2}$ $AB=AC=\sqrt{26}$ hence isosceles	M1 A1*cso [2]
(b)	Midpoint $M=(6,5)$ or gradient of $AM=1$ e.g. $y-2=\left[\frac{5-2}{6-3}\right](x-3)$ or $y-2=\text{"1"}\times(x-3)\Rightarrow y=x-1$	B1 M1A1 [3]
(c)	Equation of AC $y-2=\left(\frac{7-2}{4-3}\right)(x-3)\Rightarrow[y=5x-13]$ --- Gradient of AB $m=\frac{3-2}{8-3}\left[=\frac{1}{5}\right]$ Gradient of DB $m_p=-\frac{1}{\text{'1/5'}}[=-5]$ Equation of BD $y-3=-5(x-8)\Rightarrow[y=-5x+43]$ $5x-13=\text{'}-5x+43\text{'}\Rightarrow x=\ldots.$ or $y=\ldots$ $\left(\frac{56}{10},15\right)$ oe	M1A1 M1 M1 M1A1 dM1 A1 [8]
ALT I (c) Last 6 marks	$D(x,\text{'}5x-13\text{'})$ $BD^2=(x-8)^2+(5x-13-3)^2$ or $AD^2=(x-3)^2+(5x-13-2)^2$ $26+(x-8)^2+(5x-13-3)^2=(x-3)^2+(5x-13-2)^2$ $26-16x+64-160x+256=-6x+9-150x+225\Rightarrow x=\ldots$ $\left(\frac{56}{10},15\right)$ oe	M1 M1 M1A1 dM1 A1

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 6 of 7 · Question 2 part c · 8 marks

2025 JUN · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

ALT II (c)	$y-2=\left(\frac{7-2}{4-3}\right)(x-3)\Rightarrow[y=5x-13]$	
	or $\overrightarrow{AC}=\begin{pmatrix}1\\5\end{pmatrix}$ oe	M1A1
	$\left('4\sqrt{2}'\right)^2='\sqrt{26}'^2+'\sqrt{26}'^2-2\times'\sqrt{26}'\times'\sqrt{26}'\cos\angle BAC\Rightarrow\cos\angle BAC=...$	M1
	or $\sin\frac{\angle BAC}{2}=\frac{'2\sqrt{2}'}{'\sqrt{26}'}$	
	$\cos\angle BAC=\frac{5}{13}$ or $\angle BAC=67.4$(awrt)	M1
	$\frac{\sqrt{26}}{k\sqrt{26}}=\frac{5}{13}$ or $\cos 67.38...=\frac{\sqrt{26}}{k\sqrt{26}}$	M1
	$k=2.6$ exact	A1
	$3+1\times'2.6'=...$ and $2+5\times'2.6'=....$ or $\sqrt{(x-3)^2+('5x-13'-2)^2}='2.6\sqrt{26}'\Rightarrow x=...,y=...$	dM1
	$(5.6,15)$	A1
		Total 13 marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 6 of 7 · Question 2 part c · 8 marks

2025 JUN · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correct use of Pythagoras' theorem to find the length / length^2 of one of the sides. Must have a subtraction of x coordinates and a subtraction of y coordinates with appropriate substitution into Pythagoras' theorem. Or at least one correct vector for AC, AB, BC , or the other way around
	A1	For showing **both** AB and AC have length $=\sqrt{26}$ or length$^2 = 26$ with conclusion that the triangle is isosceles. e.g. stating AC=AB Correct vectors for AB and AC, followed by $\lvert AC\rvert = \lvert AB\rvert$ oe Allow statements such as 'hence shown' or symbols to indicate conclusion of the demonstration such as QED, // or □.
(b)	B1	For finding the midpoint M of line BC Or finds the correct gradient of $AM = 1$
	M1	For finding the equation of the line AM in any form using any correct valid method: e.g. $\begin{cases} 3m+c=2 \\ '6'm+c='5' \end{cases} \Rightarrow m=...,c=... \Rightarrow y=...$
	A1	For $y = x - 1$
(c)	M1	For correct method of finding the equation of line AC
	A1	For the correct equation of AC.
	M1	For attempting to find the gradient of AB
	M1	For negative reciprocal of **their** gradient AB
	M1	For attempting to find the equation of the line perpendicular to AB and passing through B, their gradient of BD might not be $-\frac{1}{m_{AB}}$, but must be a changed gradient
	A1	For the correct equation of BD.
	dM1	For attempting to solve their line BD and AC simultaneously. Minimum required for a correct method is to obtain an equation in either x or y (by elimination method or substitution method) and finds a value for x or y Dependent on the previous M mark only
	A1	For (5.6, 15) oe, accept given as $x = 5.6$ $y = 15$
ALT I (c) Last 6 marks	M1	Use the fact that D is on the line AC, sets up the coordinate of $D(x, '5x-13')$ or $D(x, y)$ where y is replaced by their $'5x-13'$ later
	M1	For correct attempt to find the length/length^2 expression of BD or AD using $D(x, '5x-13')$ or $D(x, y)$
	M1	For correct use of Pythagoras' theorem to set up equation $\lvert AB\rvert^2 + \lvert BD\rvert^2 = \lvert AD\rvert^2$ in terms of x only
	A1	Correct equation, simplified/ unsimplified
	dM1	For expanding and solving their equation to find x =… Dependent on the previous M mark only
	A1	For (5.6, 15) oe accept given as $x = 5.6$ $y = 15$

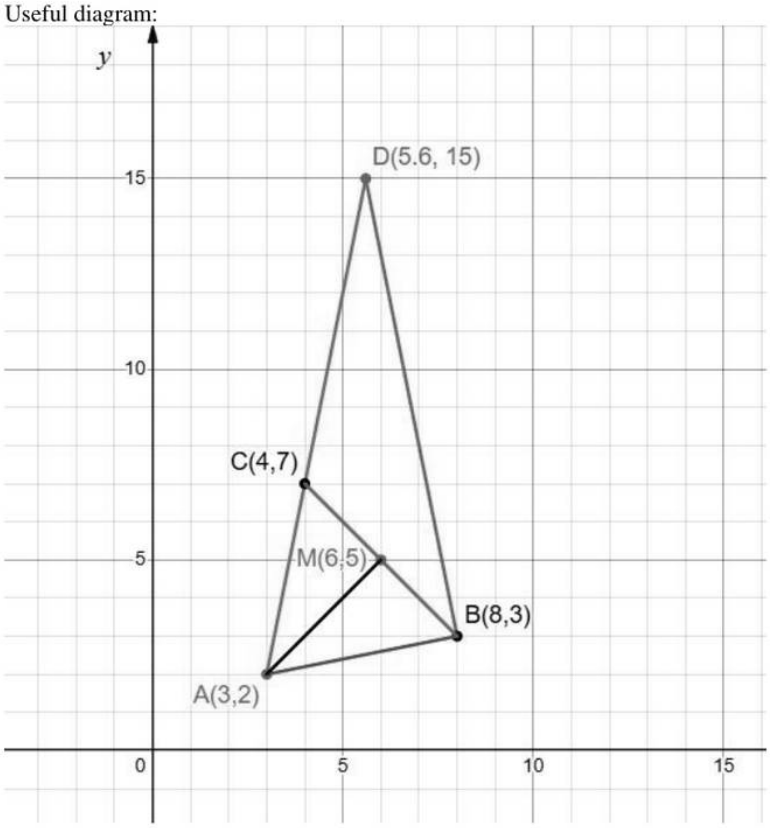
Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 6 of 7 · Question 2 part c · 8 marks

2025 JUN · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

ALT II (c)	M1	Correct method of finding the equation of line AC or Vector AC
	A1	Correct equation of AC or correct vector AC
	M1	Correct use of cosine rule or trigonometry to set up an equation involving angle BAC
	M1	Obtains a value of $\cos\angle BAC$ or $\angle BAC$ from a correct cosine rule or trigonometry equation
	M1	Correct equation in k
	A1	Correct k value, must be exact/rounded to 2.6
	dM1	Uses their k value to find $x =\ldots$ and $y =\ldots$ Dependent on the previous M mark only
	A1	Correct coordinates from an exact value of k found, not a rounded value

Useful diagram:



Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 6 of 7 · Question 2 part c · 8 marks

2025 JUN · Paper 01 · extract 5 of 5 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 7 of 7 · Question 9 part b · 5 marks

2022 JUN · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

9 a	$L_1 : y-7=m(x-4)$	B1
	Gradient of $L_2 = -\frac{1}{m}$	M1
	$L_2 : y-k=-\frac{1}{m}(x-4)$	M1
	$x=\frac{y-7}{m}+4$	M1
	$y-k=-\frac{1}{m}\left(\frac{y-7}{m}+4-4\right)$	M1
	$m^2y-m^2k=7-y$	
	$y\left(m^2+1\right)=7+m^2k$	dM1
	$Y=\frac{7+m^2k}{m^2+1}$ *	A1 cso (7)
b	Midpoint of $AB=\left([4],\frac{k+7}{2}\right)$	B1
	So $\frac{k+7}{2}=\frac{7+m^2k}{m^2+1}$	M1
	$m^2k+7m^2+k+7=2m^2k+14$	
	$m^2\left(k-7\right)=k-7$	M1
	As $k\neq 7$ $m=\pm 1$	A1
	As $m<0$ $m=-1$	A1 (5)
		Total 12 marks

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 7 of 7 · Question 9 part b · 5 marks

2022 JUN · Paper 02 · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	**NOTE** ; The algebra must be correct for the first three marks.	
	B1	For writing an equation for L_1 e.g., $y-7=m(x-4)$ **OR** Uses $y=mx+c \Rightarrow y=mx+(7-4m)$
	M1	For the gradient of the perpendicular $\left(-\frac{1}{m}\right)$
	M1	For writing an equation for L_2 (ft a changed gradient) e.g., $y-k=-\frac{1}{m}(x-4)$ **OR** Uses $y=mx+c \Rightarrow y=-\frac{1}{m}x+\left(k+\frac{4}{m}\right) \Rightarrow \left[y=\frac{-x+km+4}{m}\right]$
	M1	For rearranging their equation of L_2 to make x the subject e.g., $x=\frac{y-7}{m}+4$ or $x=km-ym+4$ **ALT 1** Rearranges either equation to make $(x-4)$ the subject $x-4=\frac{y-7}{m}$ and/or $x-4=ym+km$ **ALT 2** Eliminates y from their equations to obtain $'x-4m+7'='\frac{-x+4+km}{m}' \Rightarrow x='\frac{4+km+4m^2-7m}{m^2+1}'$ They must reach $x=\ldots\ldots$ for the award of this mark. **Allow sign errors/slips at this stage.**
	M1	Forming a linear equation in y No simplification is required at this stage. $y-k=-\frac{1}{m}\left(\frac{y-7}{m}+4-4\right)$ **ALT 1** $\frac{y-7}{m}=ym+km$ **ALT 2** Forms a linear equation in y using their expression for x by substituting back into either L_1 or L_2 **Allow sign errors/slips at this stage.**
	dM1	For attempting to rearrange to obtain the required result. **Allow sign errors but there must be no missing terms. You must check carefully.**
	A1	For the required result with no errors seen. $Y=\frac{7+m^2k}{m^2+1}$ allow $y=\frac{7+m^2k}{m^2+1}$

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 7 of 7 · Question 9 part b · 5 marks

2022 JUN · Paper 02 · extract 3 of 4 · find part b in this mark scheme extract

(b)	**B1**	For identifying y coordinate of the midpoint of AB $\left([4], \frac{k+7}{2}\right)$ the x coordinate is not required for this mark.
	M1	Equating the y coordinate of C and y coordinate of the midpoint to give $\frac{k+7}{2} = \frac{7+m^2k}{m^2+1}$
	M1	For rearranging and factorising to attempt achieve $m^2(k-7) = k-7$ This mark can implied by the correct value for either m^2 or m
	A1	Concludes that $m = \pm 1$ $\left[m^2 = \frac{k-7}{k-7} = 1 \Rightarrow m = \pm 1\right]$ Again, this can be implied by the correct $m = -1$
	A1	For $m = -1$
	ALT 1	
	B1	Let coordinates of C be (X, Y) Sets: length AC = length BC $(X-4)^2 + (Y-7)^2 = (X-4)^2 + (Y-k)^2 \Rightarrow \left[(Y-7)^2 = (Y-k)^2\right]$
	M1	Using the coordinates for C of (X, Y) or otherwise, performs the following algebraic manipulation. $(Y-7)^2 = (Y-k)^2 \Rightarrow Y^2 - 14Y + 49 = Y^2 - 2kY + k^2 \Rightarrow -14Y + 49 = -2kY + k^2$ $\Rightarrow 2Y(k-7) = (k+7)(k-7)$ $\Rightarrow 2Y = k+7$ **NB: Allow sign slips, but not missing terms**
	M1	Substitutes in the given expression for Y and attempts to obtain $m^2(k-7) = k-7$ $\frac{7+m^2k}{m^2+1} = \frac{k+7}{2}$ $\Rightarrow 14 + 2m^2k = km^2 + 7m^2 + k + 7 \Rightarrow 7 + km^2 = 7m^2 + k$ $\Rightarrow m^2(k-7) = k-7$ **NB: Allow sign slips, but not missing terms**
	A1	Concludes that $m = \pm 1$ $m^2 = \frac{k-7}{k-7} = 1 \Rightarrow m = \pm 1$
	A1	For $m = -1$
	ALT 2	
	If candidates deduces that $BAC = 45°$ and concludes that the gradient must be $m = -1$ award B1M1M1A1A1 If they leave their answer as $m = \pm 1$ or just $m = 1$, withhold the final A mark	

Type MS · Parallel and perpendicular lines: proving it or finding an unknown

Answer 7 of 7 · Question 9 part b · 5 marks

2022 JUN · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks
	$\log_2 x$ $\log_2 x$	